

WET LAB REAGENT LOGBOOK

Reagent Preparation, Mass Balance & Quality Control Tracking



REAG-LOG-2026-B3

Audit Log Ref: AL-86392-X

 Facility: Apex BioLabs Core Facility

 Unit: Buffer & Media Prep Suite B-304

TOTAL BATCHES LOGGED

142

 Quarter Q3 2026

QC APPROVED & ACTIVE

128

✓ Ready for Assay Use

PENDING QC VERIFY

9

 Hold for Sterility

EXPIRED / QUARANTINED

5

 Flagged Disposal

SECTION 1: MASTER REAGENT PREPARATION RECORD – BATCH # REAG-2026-0891

Reagent Name: 10X Tris-EDTA (TE) Buffer pH 8.0

Batch / Lot ID: REAG-2026-0891-TE80

Target Volume: 1000.0 mL (1.0 L)

Prepared By: Dr. Eleanor Vance (Analyst)

SOP Reference: SOP-BUFF-012 Rev 4.0

Preparation Date: 2026-07-24 Time: 09:30

Expiration Date: 2027-01-24 (6 Months)

Storage Condition: Room Temp (20°C to 25°C)

Verified By: Marcus Chen (QA Lead)

Quality Grade: Molecular Biology Grade

Formulation & Chemical Mass Calculation Block

Chemical Component	CAS #	MW	Target Conc.	Calc. Mass	Weighed Mass	Vendor & Lot #
Tris Base	77-86-1	121.14	100 mM	12.114 g	12.115 g	Synapse #SY-88291
[Tris(hydroxymethyl)aminomethane]						
EDTA Disodium Salt Dihydrate	6381-92-6	372.24	10 mM	3.722 g	3.723 g	Nexa #NX-44012
Hydrochloric Acid (HCl, 6M Stock)	7647-01-0	N/A	pH Adj.	~ 4.2 mL	4.18 mL	Apex #AP-90112
Ultrapure Water (Milli-Q 18.2 MΩ · cm)	7732-18-5	18.02	Diluent	Q.S. 1000 mL	982.5 g	In-House #TANK-02

Step-by-Step Preparation Protocol Execution

Step	Procedure Specification	Check	Observed Parameter / Notes
01	Tare analytical balance (Bal-04). Weigh Tris Base (12.114 g) and EDTA (3.722 g).	[✓]	Weighed 12.115 g Tris, 3.723 g EDTA.
02	Transfer solutes into 1000 mL glass beaker. Add 800 mL ultrapure ddH ₂ O.	[✓]	Dissolved under magnetic stirring at 400 RPM.
03	Calibrate pH meter (pH-02) using pH 4.01, 7.00, and 10.01 standard buffers.	[✓]	Slope = 99.2%, Temp compensation ON (22.4°C).
04	Slowly titrate with 6M HCl under stirring until target pH 8.00 ± 0.05 is achieved.	[✓]	Final titration volume: 4.18 mL 6M HCl. pH = 8.01.
05	Transfer solution to 1000 mL volumetric flask. Q.S. to mark with ddH ₂ O.	[✓]	Meniscus aligned precisely at 1000.0 mL mark.
06	Filter sterilize through 0.22 μm PES membrane filter into sterile storage bottle.	[✓]	Autoclave indicator strip turned black. Cycle #402.

Quality Control, Verification & Sign-Off

Final Measured pH:	8.01 at 22.5°C (Spec: 8.00 ± 0.05)	Conductivity:	14.2 mS/cm at 23.0°C
Physical Appearance:	Clear, colorless, particle-free	Sterility Check:	Passed (0.22 μm Filtered + Autoclaved)
DNase/RNase Test:	Negative (Fluorescence Assay Pass)	Refractometer Index:	1.3342 ± 0.0005
Analyst Signature:	Dr. Eleanor Vance	QA Approval:	Marcus Chen, Lead QA

REAGENT BATCH IDENTIFIER BARCODE LOG



REAG-2026-0891-TE80

Scan barcode for automated LIMS inventory entry and lot genealogy tracking.